

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 October 2021 Worked Solutions

October 2021 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com

P2

PAST PAPER

WMA12/01 October 2021

October 2021 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Binomial Expansion

1. The first three terms, in ascending powers of x , of the binomial expansion of $(1 + kx)^{16}$ are

$$1, -4x \text{ and } px^2$$

where k and p are constants.

- (a) Find, in simplest form,

(i) the value of k

(ii) the value of p

(3)

$$g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$$

Using the value of k found in part (a),

- (b) find the term in x^2 in the expansion of $g(x)$.

(3)

Worked Solution - Question 1

1. Compare the linear term

In $(1 + kx)^{16}$, the coefficient of x is $16k$.

2. Find k

Since the coefficient of x is -4 , $16k = -4$, so $k = -\frac{1}{4}$.

3. Find p

The coefficient of x^2 is $\binom{16}{2}k^2 = 120 \left(\frac{1}{16}\right) = \frac{15}{2}$.

4. Find the needed binomial terms

For $k = -\frac{1}{4}$, the x^2 coefficient in $(1 + kx)^{16}$ is $\frac{15}{2}$ and the x^3 coefficient is $\binom{16}{3} \left(-\frac{1}{4}\right)^3 = -\frac{35}{4}$.

5. Build the x squared term in g

In $g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$, the x^2 term comes from $2 \left(\frac{15}{2}x^2\right)$ and $\frac{16}{x} \left(-\frac{35}{4}x^3\right)$.

6. Simplify

The required term is $15x^2 - 140x^2 = -125x^2$.

Final answer

$$(a)(i) k = -\frac{1}{4}. \quad (a)(ii) p = \frac{15}{2}. \quad (b) -125x^2.$$

Question 2

Sequences & Series

2. A sequence is defined by

$$u_1 = 6$$
$$u_{n+1} = ku_n + 3$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for u_3

(2)

Given that $\sum_{n=1}^3 u_n = 117$

- (b) find the value of k .

(3)

Worked Solution - Question 2

Topic group

1. Find u_2

$$u_2 = ku_1 + 3 = 6k + 3.$$

2. Find u_3

$$u_3 = ku_2 + 3 = k(6k + 3) + 3 = 6k^2 + 3k + 3.$$

3. Use the given sum

$$u_1 + u_2 + u_3 = 117, \text{ so } 6 + (6k + 3) + (6k^2 + 3k + 3) = 117.$$

4. Simplify the equation

$$6k^2 + 9k + 12 = 117, \text{ hence } 2k^2 + 3k - 35 = 0.$$

5. Solve for k

$$(2k - 7)(k + 5) = 0. \text{ Since } k \text{ is positive, } k = \frac{7}{2}.$$

Final answer

$$(a) u_3 = 6k^2 + 3k + 3. \quad (b) k = \frac{7}{2}.$$

Question 3

Integration

3.

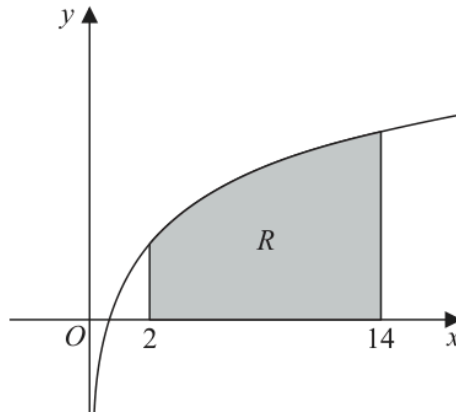


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = \log_{10} x$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 14$

Using the trapezium rule with four strips of equal width,

(a) show that the area of R is approximately 10.10 (3)

(b) Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of R . (1)

(c) Using the answer to part (a) and making your method clear, estimate the value of

(i) $\int_2^{14} \log_{10} \sqrt{x} \, dx$

(ii) $\int_2^{14} \log_{10} 100x^3 \, dx$

(4)

Worked Solution - Question 3

1. Find the strip width

Four equal strips from $x = 2$ to $x = 14$ have width $h = \frac{14 - 2}{4} = 3$.

2. List the ordinates

Use $x = 2, 5, 8, 11, 14$ and $y = \log_{10} x$.

3. Apply the trapezium rule

$$A \approx \frac{3}{2} \{ \log_{10} 2 + \log_{10} 14 + 2(\log_{10} 5 + \log_{10} 8 + \log_{10} 11) \} = 10.10 \dots$$

4. Improve the estimate

Use more strips of smaller width to make the trapezium rule estimate more accurate.

5. Estimate the square-root integral

$$\log_{10} \sqrt{x} = \frac{1}{2} \log_{10} x, \text{ so } \int_2^{14} \log_{10} \sqrt{x} dx \approx \frac{1}{2} (10.10) = 5.05.$$

6. Rewrite the second logarithm

$$\log_{10}(100x^3) = \log_{10} 100 + \log_{10} x^3 = 2 + 3 \log_{10} x.$$

7. Estimate the second integral

$$\int_2^{14} (2 + 3 \log_{10} x) dx \approx 2(12) + 3(10.10) = 54.30.$$

Final answer

(a) 10.10. (b) Use more strips.

(c)(i) 5.05. (c)(ii) 54.30.

Question 4

Polynomials

4.

$$f(x) = (x^2 - 2)(2x - 3) - 21$$

- (a) State the value of the remainder when $f(x)$ is divided by $(2x - 3)$

(1)

- (b) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$

(2)

- (c) Hence,

- (i) factorise $f(x)$

- (ii) show that the equation $f(x) = 0$ has only one real root.

(5)

Worked Solution - Question 4

1. Use the remainder theorem

Dividing by $(2x - 3)$ means use $x = \frac{3}{2}$.

2. Find the remainder

$$f\left(\frac{3}{2}\right) = \left(\frac{9}{4} - 2\right)(0) - 21 = -21.$$

3. Check x equals three

$$f(3) = (9 - 2)(6 - 3) - 21 = 7 \cdot 3 - 21 = 0.$$

4. Use the factor theorem

Since $f(3) = 0$, $(x - 3)$ is a factor of $f(x)$.

5. Expand f

$$f(x) = (x^2 - 2)(2x - 3) - 21 = 2x^3 - 3x^2 - 4x - 15.$$

6. Factorise

Dividing by $(x - 3)$ gives $2x^2 + 3x + 5$, so $f(x) = (x - 3)(2x^2 + 3x + 5)$.

7. Check the quadratic roots

For $2x^2 + 3x + 5$, the discriminant is $3^2 - 4(2)(5) = 9 - 40 = -31 < 0$.

8. Conclude

The quadratic has no real roots, so $f(x) = 0$ has only one real root, $x = 3$.

Final answer

(a) -21 . (b) Proven.

(c)(i) $f(x) = (x - 3)(2x^2 + 3x + 5)$. (c)(ii) Only real root: $x = 3$.

Question 5

Arithmetic Sequences

5. A company that owned a silver mine

- extracted 480 tonnes of silver from the mine in year 1
- extracted 465 tonnes of silver from the mine in year 2
- extracted 450 tonnes of silver from the mine in year 3

and so on, forming an arithmetic sequence.

(a) Find the mass of silver extracted in year 14

(2)

After a total of 7770 tonnes of silver was extracted, the company stopped mining.

Given that this occurred at the end of year N ,

(b) show that

$$N^2 - 65N + 1036 = 0$$

(3)

(c) Hence, state the value of N .

(1)

Worked Solution - Question 5

1. Identify the arithmetic sequence

The first term is 480 and the common difference is $465 - 480 = -15$.

2. Find year fourteen

$u_{14} = 480 + 13(-15) = 285$, so 285 tonnes were extracted in year 14.

3. Set up the total

The total after N years is $S_N = \frac{N}{2} \{2(480) + (N - 1)(-15)\}$.

4. Use the total extraction

$$\frac{N}{2}(975 - 15N) = 7770.$$

5. Show the quadratic

Multiplying by 2 and rearranging gives $15N^2 - 975N + 15540 = 0$, so $N^2 - 65N + 1036 = 0$.

6. Solve and choose N

$N = 28$ or 37 . Since extraction amounts would become negative before year 37, the stopping year is $N = 28$.

Final answer

(a) 285 tonnes.

(b) Proven: $N^2 - 65N + 1036 = 0$.

(c) $N = 28$.

Question 6

Circles

6. (i) The circle C_1 has equation

$$x^2 + y^2 + 10x - 12y = k \quad \text{where } k \text{ is a constant}$$

- (a) Find the coordinates of the centre of C_1 (2)

- (b) State the possible range in values for k . (2)

- (ii) The point $P(p, 0)$, the point $Q(-2, 10)$ and the point $R(8, -14)$ lie on a different circle, C_2

Given that

- p is a positive constant
- QR is a diameter of C_2

find the exact value of p . (4)

Worked Solution - Question 6

1. Complete the square

$$x^2 + 10x = (x + 5)^2 - 25 \text{ and } y^2 - 12y = (y - 6)^2 - 36.$$

2. Write centre-radius form

$$\text{The circle becomes } (x + 5)^2 + (y - 6)^2 = k + 61.$$

3. Read the centre

The centre is $(-5, 6)$.

4. State the range for k

For a circle with positive radius, $k + 61 > 0$, so $k > -61$.

5. Use the diameter property

If QR is a diameter and P lies on the circle, then $\angle QPR = 90^\circ$.

6. Use perpendicular vectors

So $(Q - P) \cdot (R - P) = 0$. With $P = (p, 0)$, $Q = (-2, 10)$, and $R = (8, -14)$, this gives $(-2 - p, 10) \cdot (8 - p, -14) = 0$.

7. Form the quadratic

$$(-2 - p)(8 - p) - 140 = 0, \text{ so } p^2 - 6p - 156 = 0.$$

8. Choose the positive value

$$p = \frac{6 \pm \sqrt{660}}{2} = 3 \pm \sqrt{165}. \text{ Since } p \text{ is positive, } p = 3 + \sqrt{165}.$$

Final answer

(i)(a) $(-5, 6)$. *(i)(b)* $k > -61$. *(ii)* $p = 3 + \sqrt{165}$.

Question 7

Geometric Sequences

7. (i) A geometric sequence has first term 4 and common ratio 6
Given that the n^{th} term is greater than 10^{100} , find the minimum possible value of n . (3)

- (ii) A different geometric sequence has first term a and common ratio r .

Given that

- the second term of the sequence is -6
- the sum to infinity of the series is 25

- (a) show that

$$25r^2 - 25r - 6 = 0 \quad (3)$$

- (b) Write down the solutions of

$$25r^2 - 25r - 6 = 0 \quad (1)$$

Hence,

- (c) state the value of r , giving a reason for your answer, (1)

- (d) find the sum of the first 4 terms of the series. (2)

Worked Solution - Question 7

1. Write the n th term

The n th term of the first sequence is $4 \cdot 6^{n-1}$.

2. Use logarithms

We need $4 \cdot 6^{n-1} > 10^{100}$, so $n - 1 > \frac{100 - \log_{10} 4}{\log_{10} 6} = 127.736 \dots$

3. Choose the smallest integer

Thus n must be greater than $128.736 \dots$, so the minimum possible value is **129**.

4. Use the second term

For the second sequence, $ar = -6$, so $a = -\frac{6}{r}$.

5. Use the sum to infinity

$\frac{a}{1-r} = 25$. Substituting $a = -\frac{6}{r}$ gives $-6 = 25r(1-r)$.

6. Show the quadratic

Rearranging gives $25r^2 - 25r - 6 = 0$.

7. Solve for r

$25r^2 - 25r - 6 = 0$ gives $r = \frac{6}{5}$ or $r = -\frac{1}{5}$.

8. Choose the valid ratio

A sum to infinity exists only when $|r| < 1$, so $r = -\frac{1}{5}$.

9. Find the first term

Since $ar = -6$, $a = 30$.

10. Sum four terms

$$S_4 = 30 \frac{1 - (-1/5)^4}{1 - (-1/5)} = \frac{624}{25}.$$

Final answer

(i) $n = 129$. (ii)(a) Proven.

(ii)(b) $r = \frac{6}{5}$ or $-\frac{1}{5}$. (ii)(c) $r = -\frac{1}{5}$. (ii)(d) $\frac{624}{25}$.

Question 8

Applications of Differentiation

8. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

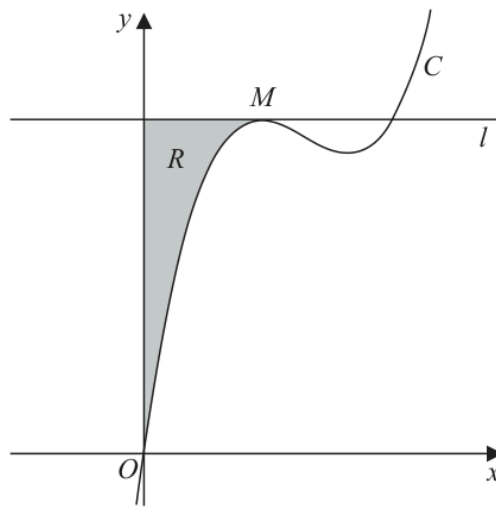


Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{4}{3}x^3 - 11x^2 + kx \quad \text{where } k \text{ is a constant}$$

The point M is the maximum turning point of C and is shown in Figure 2.

Given that the x coordinate of M is 2

- (a) show that $k = 28$ (3)

- (b) Determine the range of values of x for which y is increasing. (2)

The line l passes through M and is parallel to the x -axis.

The region R , shown shaded in Figure 2, is bounded by the curve C , the line l and the y -axis.

- (c) Find, by algebraic integration, the exact area of R . (5)

Worked Solution - Question 8

Topic group

1. Differentiate the curve

For $y = \frac{4}{3}x^3 - 11x^2 + kx$, $\frac{dy}{dx} = 4x^2 - 22x + k$.

2. Use the turning point

At the turning point M , $x = 2$ and $\frac{dy}{dx} = 0$.

3. Show k

$4(2)^2 - 22(2) + k = 0$, so $16 - 44 + k = 0$ and $k = 28$.

4. Factor the derivative

With $k = 28$, $\frac{dy}{dx} = 4x^2 - 22x + 28 = 2(2x - 7)(x - 2)$.

5. Find where y is increasing

The derivative is positive outside its roots, so y is increasing for $x < 2$ or $x > \frac{7}{2}$.

6. Find the height of the line

At $x = 2$, $y = \frac{4}{3}(8) - 11(4) + 28(2) = \frac{68}{3}$. Hence line l has equation $y = \frac{68}{3}$.

7. Set up the shaded area

The region is between the line and the curve from $x = 0$ to $x = 2$, so

$$A = \int_0^2 \left(\frac{68}{3} - \left(\frac{4}{3}x^3 - 11x^2 + 28x \right) \right) dx.$$

8. Integrate

$$A = \left[\frac{68}{3}x - \frac{1}{3}x^4 + \frac{11}{3}x^3 - 14x^2 \right]_0^2.$$

9. Evaluate exactly

Substituting $x = 2$ gives $\frac{136}{3} - \frac{16}{3} + \frac{88}{3} - 56 = \frac{40}{3}$.

Final answer

(a) Proven: $k = 28$.

(b) $x < 2$ or $x > \frac{7}{2}$. (c) $\frac{40}{3}$.

Question 9

Proof

9. (a) Prove that for all positive values of x and y ,

$$\frac{x+y}{2} \geq \sqrt{xy} \quad (3)$$

- (b) Prove by counter-example that this inequality does not hold when x and y are both negative.

(1)

Worked Solution - Question 9

Topic group

1. Start with a square

For positive x and y , both $\sqrt{x}$ and $\sqrt{y}$ are real, so $(\sqrt{x} - \sqrt{y})^2 \geq 0$.

2. Expand

$$(\sqrt{x} - \sqrt{y})^2 = x - 2\sqrt{xy} + y \geq 0.$$

3. Rearrange

Thus $x + y \geq 2\sqrt{xy}$, and dividing by 2 gives $\frac{x + y}{2} \geq \sqrt{xy}$.

4. Give a negative counterexample

Take $x = -1$ and $y = -1$. Then $\frac{x + y}{2} = -1$, while $\sqrt{xy} = \sqrt{1} = 1$.

5. Conclude

Since $-1 \geq 1$ is false, the inequality does not hold for all negative x and y .

Final answer

(a) Proven.

(b) For example, $x = y = -1$ gives $-1 \not\geq 1$.

Question 10

Trigonometric Equations

10. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3 \quad (5)$$

- (ii) Solve, for $0 < \theta < 360^\circ$

$$(2 \sin \theta - \cos \theta)^2 = 1$$

giving your answers, as appropriate, to one decimal place.

(5)

(Total 10 marks)

Worked Solution - Question 10

1. Take the square root

Let $u = 2x + \frac{\pi}{4}$. From $\tan^2 u = 3$, $\tan u = \pm\sqrt{3}$.

2. Write the angle families

Thus $u = \frac{\pi}{3} + n\pi$ or $u = -\frac{\pi}{3} + n\pi$, where n is an integer.

3. Return to x

So $x = \frac{\pi}{24} + \frac{n\pi}{2}$ or $x = -\frac{7\pi}{24} + \frac{n\pi}{2}$.

4. Use the interval

For $-\frac{\pi}{2} < x < \frac{\pi}{2}$, the solutions are $x = -\frac{11\pi}{24}, -\frac{7\pi}{24}, \frac{\pi}{24}, \frac{5\pi}{24}$.

5. Expand the squared equation

$(2 \sin \theta - \cos \theta)^2 = 1$ gives $4 \sin^2 \theta - 4 \sin \theta \cos \theta + \cos^2 \theta = 1$.

6. Use the identity

Since $1 = \sin^2 \theta + \cos^2 \theta$, this simplifies to $3 \sin^2 \theta - 4 \sin \theta \cos \theta = 0$.

7. Factor

$\sin \theta (3 \sin \theta - 4 \cos \theta) = 0$.

8. Solve the two cases

$\sin \theta = 0$ gives $\theta = 180^\circ$ in the interval. Also $3 \sin \theta = 4 \cos \theta$ gives $\tan \theta = \frac{4}{3}$.

9. State theta

For $0 < \theta < 360^\circ$, $\tan \theta = \frac{4}{3}$ gives $\theta = 53.1^\circ$ and 233.1° . Therefore $\theta = 53.1^\circ, 180^\circ, 233.1^\circ$.

Final answer

$$(i) x = -\frac{11\pi}{24}, -\frac{7\pi}{24}, \frac{\pi}{24}, \frac{5\pi}{24}. \quad (ii) \theta = 53.1^\circ, 180^\circ, 233.1^\circ.$$